

WPT System budget

Inverter circuit

Component	Quantity	Price each	Taxes (EGP)	Shipping (EGP)	Total price (EGP)
G3R40MT12D SiC Mosfet	4	131.91	75	290.19	1025
Aluminum Heatsink 100*150*20	1	260	—	70	330
PCB Prototype Board	1	18	—	—	18
PCB for Inverter	1	800	—	100	900
STM32F401 microcontroller	1	320	—	—	320
ST-LINK V2 Emulator	1	220	—	—	220
3.3V voltage adapter	1	86	—	—	86
IR2110 Gate driver	2	95	—	70	260
UF4007 Fast recovery diode	6	1	—	—	6
Carbon Resistor 13Ω 0.25W	10	0.15	—	—	1.5
22uF 160V Capacitor	2	1.5	—	—	3
0.22uF 50V Capacitor	3	0.75	—	—	2.25
490nF 275V Ceramic Capacitor	4	2	—	—	8
Breadboard	1	40	—	—	40
5.6k 0.25W ohm Resistor	10	0.15	—	—	1.5
6N135 Optocoupler	2	30	—	—	60
220V to 12Vdc converter	3	65	—	—	195
Terminal block 2 pins	5	3.25	—	—	16.25
Red Led	1	0.5	—	—	0.5
Jumper Wire	30	0.75	—	—	22.5
Total Price:					3515.5

Primary Compensation

Component	Quantity	Price each	Taxes (EGP)	Shipping (EGP)	Total price (EGP)
0.33uF Resonance Film Capacitor	14	40	—	70	630
PCB prototype board	1	32	—	—	32
Total Price:					662

Induction Coils

Component	Quantity	Price each	Taxes (EGP)	Shipping (EGP)	Total price (EGP)
Litz Wire 0.1*600 strands (20m)	2	2100	4,477.94	7000	15677.94
Iclyric Board 65*65cm	4	822.5	—	—	3290
Iclyric Rod 20cm	4	167.5	—	—	670
PC40 Mn-Zn Soft Ferrite core 4 pieces 60*16*8	6	425	1580	Free	4130
Total Price:					23767.94

Secondary Compensation

Component	Quantity	Price each	Taxes (EGP)	Shipping (EGP)	Total price (EGP)
0.33uF Resonance Film Capacitor	14	40	—	70	630
PCB prototype board	1	32	—	—	32
Total Price:					662

Secondary Circuit

Component	Quantity	Price each	Taxes (EGP)	Shipping (EGP)	Total price (EGP)
220uF 450V Capacitor	3	35	—	70	175
SEI60-10A Diode 1000V 60A	4	90	—	—	360
Secondary Rectifier PCB	1	200	—	—	200
5.8ohm 5KW Load	1	2000	—	20	2020
Total Price:					2755

Other

Component	Quantity	Price each	Taxes (EGP)	Shipping (EGP)	Total price (EGP)
Electrical Box PI65 (30*40*17)	1	1280	—	—	1280
Electrical Box PI65 (25*35*15)	1	680	—	—	680
DC Fan 12V 0.22A	3	55	—	—	165
Male Blade crimp Terminal	100	1	—	—	100
Female Blade crimp Terminal	100	1	—	—	100
AC Power Socket 3 Pin	1	15	—	—	15
PCB Male Terminals	20	1	—	—	20
Stranded 6mm copper wire (5m)	2	50	—	—	100
Total Price:					2460

Total Production Price for The Project

Part	Total price (EGP)
Inverter Circuit	3515.5
Primary Compensation	662
Induction Coils	23767.94
Secondary Compensation	662
Secondary Circuit	2755
Other	2460
Total Price:	33822.44